

BENJAMIN LAM

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Work Experience

Tesla, Inc.

Jan 2025 – Present

Drive Unit Test Engineer

- **Established processes for drive inverter fault-response testing**, authoring the first test specifications and procedures and enabling systematic expansion of this test category. Work cross functionally with firmware, safety, motor controls, and electrical board design teams
- **Developing a real-time fault detection system** for all lab test stations to preserve critical teardown data and reduce repeat testing after catastrophic failure. **Employed machine learning methods (Python)** to account for large variability across test profiles and motor types. **Integrating into existing Python/NI Veristand test stand codebase**
- Characterized inverter electrical behavior during fault events, evaluating gate drive signals, phase currents, and protection timing to inform drive unit replacement decisions and mitigate damage. **Automated test execution and data acquisition (Oscilloscope automation, Python)**, achieving ~10× faster execution and improved data traceability
- Evaluated edge-case electrical readiness of inverter design changes, assessing worst-case current and protection margins. Identified reliability risks in cost-down modifications and drove mitigation prioritization, validation testing, and risk assessment
- Managing testing operations for a major drive unit program, responsible for test planning, coordination with design teams, execution, test setup assembly, and analysis to support change validation and failure analysis
- Owned root cause analysis of a critical component failure, creating new electrical and thermal metrics to evaluate hardware and firmware solutions and recommending a firmware fix deployed fleetwide, improving system safety and reliability

Ceder Group (UC Berkeley)

Aug 2022 – Dec 2024

Graduate Student Researcher (Battery Materials)

- Designed + built **automated in-situ XRD sample loader robot** (Flask/Python/I²C) enabling high-throughput characterization for integration into our lab's autonomous synthesis lab [\[Github\]](#)
- Designed + built **custom Li-air battery test system** controlled with Flask/Python/I²C to cycle cells in PID-controlled relative humidity/gas composition. Designed Swagelok-type cell for airtight sealing, assembly/control of electronic sensors and actuators, and all software. **Added Li-air battery test capabilities to our lab** (Python, C++, JavaScript, HTML) [\[Github\]](#)
- Identified the root-cause failure mechanism of Li-NASICON solid-state conductors in humid air battery cells, **eliminating all phosphate chemistries from consideration due to inherent instability in alkaline conditions**. Conducted degradation research using time-resolved XRD, SEM/EDS, and ICP-OES, with data analysis in MATLAB. Experimental results supported by DFT surface simulations and ab-initio Pourbaix diagrams. [\[Advanced Energy Materials \(2024\), First Author\]](#)

Projects

- *Drone Project (Ongoing)*: Writing flight control firmware for the nRF52840 (**Nordic Connect SDK/Zephyr RTOS in C**), implementing real-time sensor fusion (I²C accel/gyro) and Bluetooth wireless control
- *Microfluidics PCB*: Developed a fully integrated solenoid array controller for microfluidic chip testing, with 32 parallel solenoid drivers, real-time pressure sensing, on-board power conversion, and CAN/I²C communication.

Education

University of California, Berkeley

Dec 2024

MS in Materials Science Engineering, **GPA: 3.97**

- *Model Predictive Control Final Project*: **Automated path generation and in-route correction for parking** a 2-wheeled rover. Simulations with wheel slip considered to evaluate control robustness to external disturbances

University of California, Irvine

Jun 2022

BS in Mechanical Engineering and Materials Science Engineering (Double Major), **GPA: 3.98**

- Dean's Honor List (All quarters), Graduated Summa Cum Laude
- *UCI Solar Car Battery Lead*: Directed recruitment and selection of new electrical team members
- *Bowman Lab Research*: Custom MATLAB library developed for processing of EIS datasets, contributed to **2 papers**: [\[ACS Applied Nano Materials \(2023\)\]](#) [\[Advanced Functional Materials \(2024\)\]](#)
- *Hui Lab Research*: Developed a simulated model for a microfluidic valve. **Made new valve design** with truncated valve floor helps improve size variation of generated droplets (down to $\sigma = 1\%$, **3x improvement**) [\[Micromachines \(2022\)\]](#)

Skills

Electrical: **Oscilloscope, Power Analyzer**, PCB Prototyping, High voltage experience, Inverters, AC motor controls, Fault response

Programming: **Python (3+ years)**, **MATLAB (4+ years)**, C, Java (2 years)

Mechanical: **SolidWorks (CSWA Certified)**, Autodesk Fusion 360, Autodesk CFD, Dynamometer lab

Materials Science: XRD, **SEM/EDS**, ICP-OES/MS, EIS, Aqueous Chemistry, TGA/DSC, Chemical Lab Experience (3+ years)